SUPPORTING ADULT GUIDANCE

The circle and its parts

Circle Theorems · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer.
You do not need to be a mathematics specialist to support this lesson.

What this lesson does

This opening lesson establishes the vocabulary of the circle — centre, radius, diameter, chord, arc, tangent, circumference, sector, segment — and, crucially, the idea that an angle **subtends** the arc that does not contain its vertex. Precise naming is not decoration here: every theorem in the unit is stated and justified in these terms, so security with the words is the foundation the rest is built on.

Driving Question: *What are all the lines and regions hiding inside a circle?*

The mathematics, in plain terms

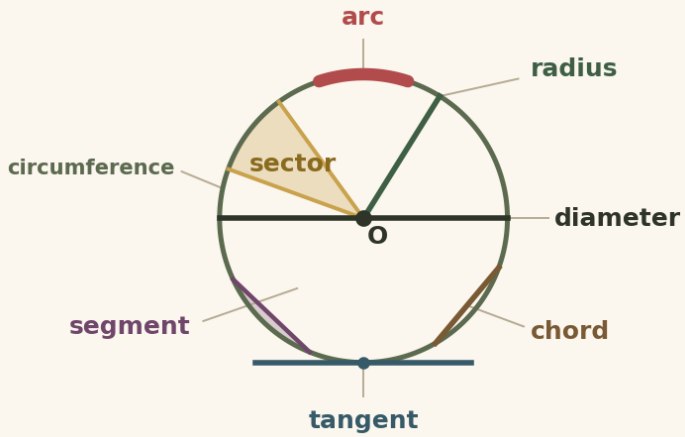
A circle is the set of points a fixed distance (the radius) from a centre; the curve is its circumference and a diameter is a chord through the centre, equal to twice the radius and the longest chord. Chords, tangents, arcs, sectors and segments are defined by what bounds them. An inscribed angle ∠APB subtends the arc AB not containing P; the same arc subtends a central angle ∠AOB — the relationship examined next lesson, and the keystone of the whole unit.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Name the parts of a circle

Identify each labelled feature and say whether it is a line, a curve or a region.

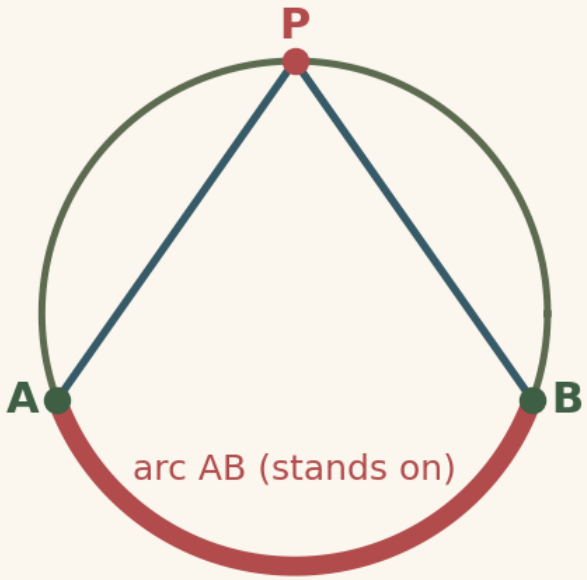


1. From the centre O out to the curve — one straight line.	radius
2. Right across through O — the longest chord.	diameter
3. Two points on the curve, not through O.	chord (a line); the curve itself is the circumference
4. A wedge between two radii, and the cap between a chord and its arc.	sector (region) and segment (region)

Lines: radius, diameter, chord, tangent. Curves: arc, circumference. Regions: sector, segment. The centre O is a single point.

Example 2 — Which arc does the angle stand on?

Points A, B and P lie on the circle. On which arc does ∠APB stand?



1. Find the vertex of the angle — the point at the corner.	the vertex is P
2. The angle stands on the arc AB on the FAR side from P.	the arc not containing P
3. Slide P along its own arc — the arc it stands on does not change.	still the same arc AB

∠APB stands on (subtends) the arc AB that does not contain P — here the lower, major arc. The same arc AB also subtends an angle at the centre, ∠AOB, which the next lesson connects to it.

Name by property, not by picture

Encourage the learner to justify each name by the property that fixes it ('from the centre to the curve → radius') rather than by how it looks, because busy exam diagrams rotate and overlap the parts. Insist on the distinction that most often slips: a chord is only a diameter if it passes through the centre, and a sector is bounded by two radii while a segment is bounded by a chord. Introduce 'stands on / subtends' carefully — it is new language that the entire unit will lean on.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Identify each part on a labelled circle, sorting lines, curves and regions.	~12 min
Movement	Match each part to its defining property.	~12 min
Reflection	Drag P and see which arc the angle ∠APB stands on.	~9 min
Creativity	Find the longest chord (the diameter) and tell a sector from a segment.	~8 min
Rest	A short pause to let the new vocabulary settle.	~2 min
Nutrition	Intellectual: the power of precise language, and anticipation of where 'stands on an arc' leads.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Calling any chord a diameter (or any line to the curve a radius).
Tie the name to the property: 'Does it pass through the centre? Only then is it a diameter. Does it start at the centre? Only then is it a radius.'

Confusing a sector with a segment.
'What bounds it? Two radii and an arc → sector (a pizza slice). One chord and an arc → segment (a slice off the top).'

Thinking the angle stands on the arc that contains P.
'The angle stands on the arc opposite its vertex — the far arc, the one without P. Point to the arc P is NOT on.'

Believing a longer chord than the diameter is possible.
'Slide a chord toward the centre — it grows, and it is longest exactly through the centre. No chord can beat the diameter.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: knowing a circle as a round curve, angle notation such as ∠APB, and the idea of a straight line joining two points. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner labels a chord through the centre as a 'chord' but not a 'diameter', or says ∠APB stands on the near arc. The aim is a question that returns the defining property.

- “Does that line pass through the centre? Then it has a more specific name — which one?”
- “What bounds this region — two radii, or a chord? So is it a sector or a segment?”
- “Where is the vertex of the angle? The arc it stands on is the one that does NOT contain that point.”
- “Slide the chord toward the centre — when is it longest? What is that longest chord called?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is a diameter a chord?

Yes — a diameter is a chord that happens to pass through the centre. It is the longest chord, and equals two radii.

What is the difference between a sector and a segment?

A sector is bounded by two radii and an arc (like a slice of pizza cut from the centre). A segment is bounded by one chord and an arc (like a slice taken off the top). The bounding lines are the tell.

What does 'subtend' mean?

An angle (or a chord) subtends — 'stands on' — an arc. The inscribed angle ∠APB subtends the arc AB that does not contain P. It is the single idea the whole unit is built on, so it is worth naming clearly now.

Why start a theorems unit with just vocabulary?

Because every theorem is stated and proved in these words. A learner secure with 'the angle subtends this arc' can follow the angle-at-the-centre theorem next lesson; a learner unsure of the words cannot. The investment here pays off across the whole unit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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